Incompressible Stokes Flow Annulus Benchmark Using Underworld3

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Abstract. We reproduce the annulus benchmark of Kramer et al. (2021) (hereafter K2021) using Underworld3 (hereafter UW3; Moresi et al., 2025). The benchmark considers incompressible, isoviscous Stokes flow in an annulus with analytical velocity and pressure solutions derived from prescribed buoyancy forcing. The benchmark includes both free-slip and zero-slip velocity boundary conditions, together with smooth and interface-localised density forcing. The UW3 calculations employ $P_2 \times P_1$ Taylor–Hood finite elements on progressively refined annulus meshes. Numerical accuracy is evaluated using relative volume L_2-norm errors for velocity and pressure. For smooth forcing cases, the solutions recover approximately second-order pressure convergence together with systematic second-order or higher velocity convergence across the tested refinement range. Cases involving delta-function forcing exhibit reduced but consistent convergence behaviour, with velocity convergence rates approaching $O(h^{1.5})$ and pressure convergence rates approaching $O(h^{0.5})$, consistent with the reduced regularity associated with interface-localised buoyancy forcing. These results verify the UW3 annulus Stokes implementation for curved-shell geometries under both free-slip and zero-slip boundary conditions considered in K2021.

1 Introduction

Finite-element modelling has become an essential tool in geodynamics for investigating mantle convection and related solid-Earth processes in geometries that more closely represent planetary interiors. In particular, cylindrical annulus and spherical-shell domains are widely used because they naturally incorporate radial gravity and curved boundaries while avoiding some of the limitations of Cartesian approximations (e.g. Baumgardner, 1985; Jarvis, 1993; Zhong et al., 2000; Tackley, 2008; Davies et al., 2022). Accurate numerical treatment of incompressible Stokes flow in such geometries requires careful consideration of mixed finite-element stability, pressure nullspaces, curved boundary representation, and boundary-condition implementation. These issues become especially important when modelling free-slip boundaries and sharp density contrasts, both of which commonly arise in geodynamical applications.

Benchmarking and verification therefore play a central role in the development and validation of geodynamical modelling software. Community benchmark studies are widely used to assess numerical accuracy, convergence behaviour, and solver robustness across different finite-element discretisations and software frameworks (e.g. Blankenbach et al., 1989; Busse et al., 1994; King et al., 2010; Tosi et al., 2015; Thieulot and Puckett, 2018; Kramer et al., 2021; Thieulot and Bangerth, 2022). Benchmarks in curved annulus geometries are particularly valuable because they provide analytical solutions that enable di-

rect evaluation of velocity, pressure, and boundary-condition accuracy in geometries relevant to regional and global mantle dynamics.

In this study, we reproduce the annulus benchmark of K2021 using UW3. The benchmark considers incompressible isoviscous Stokes flow in an annulus with analytical velocity and pressure solutions derived from prescribed buoyancy forcing. The benchmark includes both free-slip and no-slip velocity boundary conditions together with smooth and interface-localised density forcing. The smooth forcing cases assess convergence behaviour for regular analytical solutions, while the interface-localised forcing cases provide a more stringent test involving reduced solution regularity and sharp buoyancy variations across internal interfaces.

We investigate the convergence behaviour and numerical accuracy of the UW3 annulus Stokes implementation using $P_2 \times P_1$ Taylor–Hood finite elements on progressively refined annulus meshes. Numerical solutions are compared against the analytical benchmark solutions using relative L_2-norm errors for velocity and pressure. Particular attention is given to the implementation of free-slip boundary conditions and the convergence behaviour associated with smooth and interface-localised forcing cases. The results provide verification of UW3 for curved-domain annulus Stokes flow and demonstrate the effectiveness of the Kramer et al. benchmark for assessing finite-element accuracy in annulus geometries.

2 Governing Equations and Benchmark Solution

The benchmark solves the steady incompressible Stokes equations in an annulus $\Omega = \{(r,\theta) : R_- \le r \le R_+\}$:

$$-\nabla \cdot \boldsymbol{\sigma} = \rho \mathbf{g}, \tag{1a}$$

$$\nabla \cdot \mathbf{u} = 0, \tag{1b}$$

where $\mathbf{u}$ is velocity, p is pressure, ρ is density, and $\mathbf{g}$ is gravitational acceleration. The stress tensor is

$$\boldsymbol{\sigma} = -p\mathbf{I} + 2\eta\boldsymbol{\varepsilon}(\mathbf{u}), \qquad \boldsymbol{\varepsilon}(\mathbf{u}) = \frac{1}{2}\left(\nabla\mathbf{u} + \nabla\mathbf{u}^T\right), \tag{2}$$

with constant viscosity η.

Following K2021, the annulus radii are

$$R_- = 1.22, \qquad R_+ = 2.22.$$

For the interface-localised cases, the internal density interface is located at

$$R_{\text{int}} = 2.0.$$

The analytical forcing is separated into azimuthal and radial parts. Smooth cases use a density perturbation proportional to

$$\left(\frac{r}{R_+}\right)^k \cos(n\theta), \tag{3}$$

where k controls the radial structure and n controls the azimuthal wavenumber. Delta-function cases use an interface forcing proportional to

$$\delta(r - R_{\mathrm{int}})\cos(n\theta). \tag{4}$$

The full analytical velocity and pressure fields are those derived by K2021; the UW3 implementation evaluates these closed-form solutions to define the forcing, impose the benchmark boundary conditions, and measure numerical error.

3 Numerical Setup

The UW3 calculations use $P_2 \times P_1$ Taylor–Hood elements, with quadratic velocity and continuous linear pressure. This mixed pair is a standard choice for incompressible Stokes flow and is expected to provide robust pressure behaviour for smooth solutions (Girault and Raviart, 1986; Boffi et al., 2013). The pressure nullspace is removed before error evaluation because incompressible Stokes flow determines pressure only up to an additive constant.

The verification suite contains four benchmark groups (Table 1). Cases 1 and 2 use free-slip boundaries, while Cases 3 and 4 use zero-slip boundaries. Cases 1 and 3 use delta-function forcing on the internal interface; Cases 2 and 4 use smooth volumetric forcing.

Table 1. Benchmark cases used in the UW3 verification runs.

Case	Forcing	Boundary condition	Parameters
1	Delta function	Free slip	$n = 2, 8, 32$
2	Smooth	Free slip	$n = 2, 8, 32;\ k = 2, 8$
3	Delta function	Zero slip	$n = 2, 8, 32$
4	Smooth	Zero slip	$n = 2, 8, 32;\ k = 2, 8$

Figure 1 shows representative analytical density perturbations. Increasing n increases the angular complexity of the benchmark solution. For smooth forcing, increasing k changes the radial weighting of the density field. For delta-function forcing, the density is concentrated on the internal interface and therefore reduces solution regularity relative to the smooth cases.

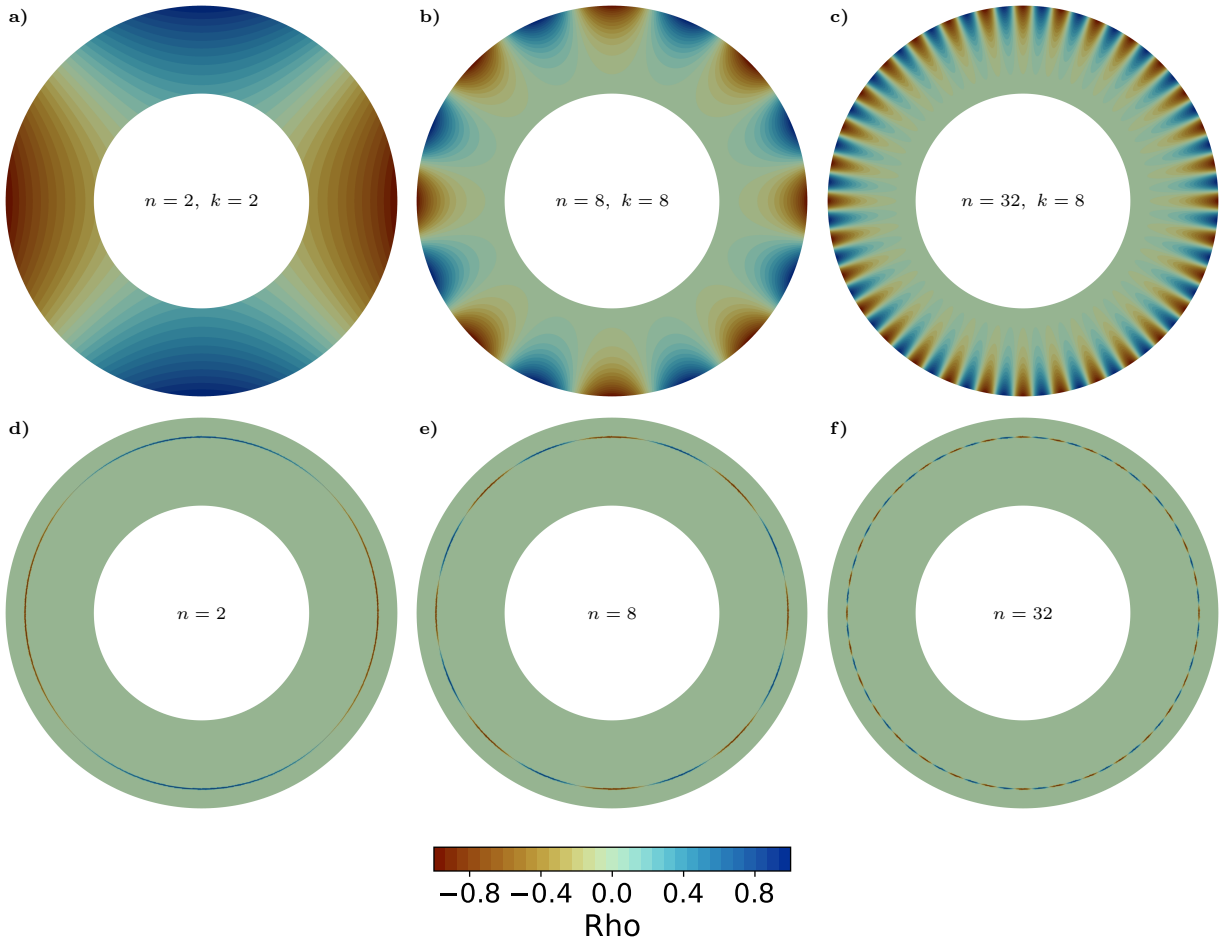

Figure 1. Analytical density perturbations for the Kramer annulus benchmark. The upper row shows smooth forcing and the lower row shows delta-function forcing. Columns correspond to increasing azimuthal complexity.

4 Error Metrics

Velocity and pressure errors are reported as relative volume L_2 norms. For a numerical field q_h and analytical reference field q^*, the relative error is

$$E_{L_2}^{\mathrm{rel}}(q) = \left(\frac{\int_\Omega |q_h - q^*|^2 \, \mathrm{d}\Omega}{\int_\Omega |q^*|^2 \, \mathrm{d}\Omega} \right)^{1/2}. \tag{5}$$

Thus the table and figure labels use $E_{L_2}^{\mathrm{rel}}(\mathbf{u})$ for velocity and $E_{L_2}^{\mathrm{rel}}(p)$ for pressure. For two consecutive refinements with characteristic mesh sizes h_1 and h_2, the observed convergence rate is

$$\alpha = \frac{\log(E_{h_1}/E_{h_2})}{\log(h_1/h_2)}. \tag{6}$$

With uniform halving of the nominal mesh size, this reduces to $\alpha = \log_2(E_{h_1}/E_{h_2})$.

5 Results

The convergence results are shown in Fig. 2. For the smooth forcing cases, pressure errors converge close to $O(h^2)$ for both free-slip and zero-slip boundary conditions. Velocity errors converge systematically and are approximately second order for $n = 2$ and $n = 8$. The high-wavenumber $n = 32$ smooth cases show larger error constants and stronger pre-asymptotic behaviour, but the rates decrease toward the expected refined-mesh trend as the solution becomes better resolved.

The delta-function forcing cases exhibit lower rates. Velocity convergence approaches $O(h^{1.5})$, while pressure convergence approaches $O(h^{0.5})$ for both free-slip and zero-slip boundaries. This behaviour is expected for the K2021 benchmark because the Green's-function solution associated with interface-localised forcing is less regular than the smooth volumetric-forcing solution. The reduced pressure rate is therefore interpreted as a property of the benchmark solution regularity rather than as a failure of the Stokes solver.

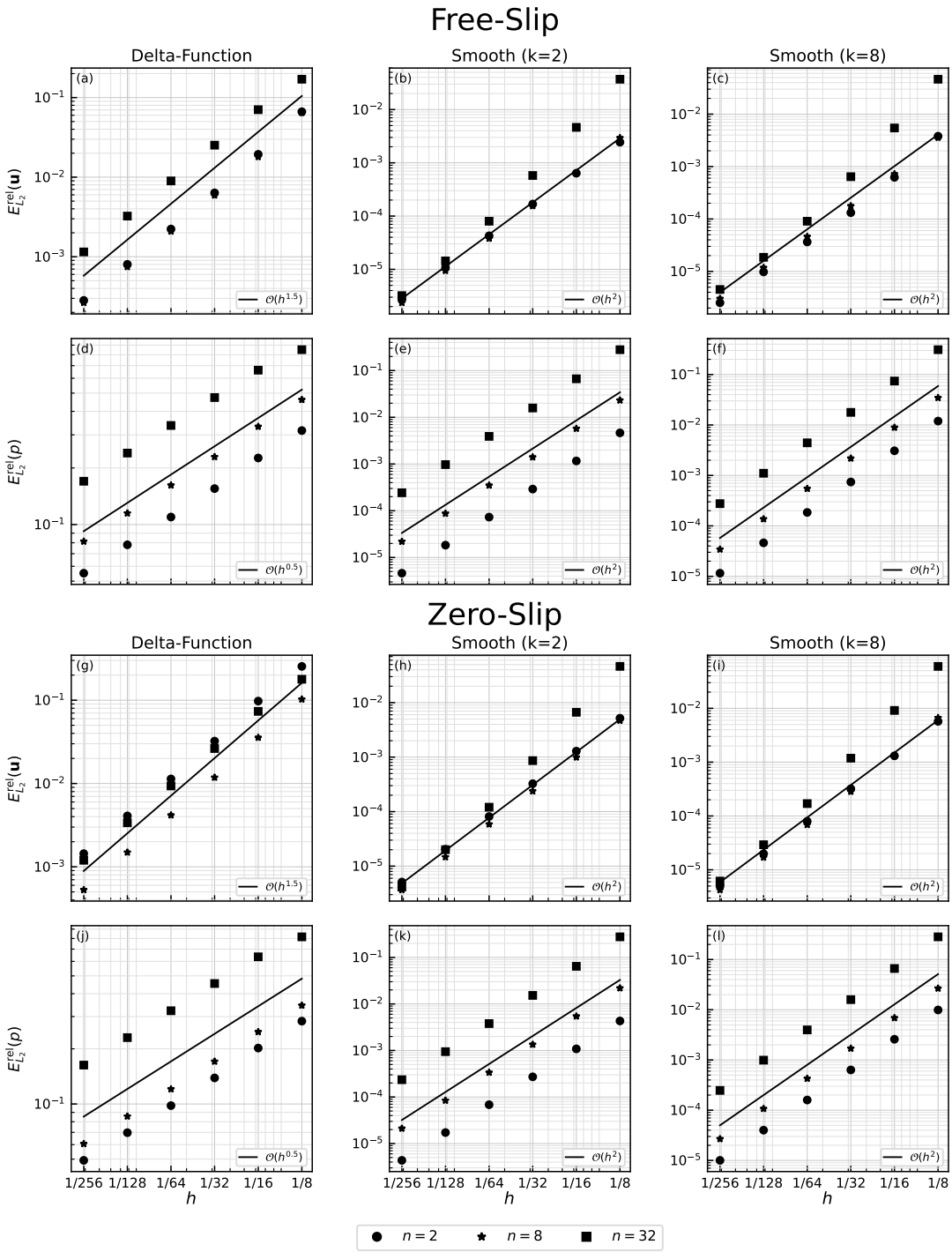

Figure 2. Velocity and pressure convergence for the four Kramer annulus cases. Columns show delta-function forcing and smooth forcing with $k = 2$ and $k = 8$. The upper block uses free-slip boundaries and the lower block uses zero-slip boundaries.

The numerical errors and pairwise convergence rates are provided in Tables 2–7. Each table groups three azimuthal wavenumbers and reports mesh size, velocity error, velocity rate, pressure error, and pressure rate.

Table 2. Free-slip, delta-function forcing.

h	$n=2$				$n=8$				$n=32$			
	$E^{\mathrm{rel}}_{L_2}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2}(p)$	rate	$E^{\mathrm{rel}}_{L_2}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2}(p)$	rate	$E^{\mathrm{rel}}_{L_2}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2}(p)$	rate
$1/8$	6.636e-02	–	3.161e-01	–	6.455e-02	–	4.610e-01	–	1.700e-01	–	8.511e-01	–
$1/16$	1.936e-02	1.78	2.260e-01	0.48	1.798e-02	1.84	3.318e-01	0.47	7.045e-02	1.27	6.624e-01	0.36
$1/32$	6.346e-03	1.61	1.553e-01	0.54	5.914e-03	1.60	2.290e-01	0.54	2.523e-02	1.48	4.736e-01	0.48
$1/64$	2.230e-03	1.51	1.097e-01	0.50	2.084e-03	1.50	1.617e-01	0.50	8.968e-03	1.49	3.364e-01	0.49
$1/128$	8.026e-04	1.47	7.804e-02	0.49	7.460e-04	1.48	1.149e-01	0.49	3.242e-03	1.47	2.400e-01	0.49
$1/256$	2.828e-04	1.50	5.512e-02	0.50	2.643e-04	1.50	8.132e-02	0.50	1.150e-03	1.50	1.699e-01	0.50

Table 3. Free-slip, smooth forcing ($k=2$).

h	$n=2$				$n=8$				$n=32$			
	$E^{\mathrm{rel}}_{L_2}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2}(p)$	rate	$E^{\mathrm{rel}}_{L_2}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2}(p)$	rate	$E^{\mathrm{rel}}_{L_2}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2}(p)$	rate
$1/8$	2.441e-03	–	4.639e-03	–	2.935e-03	–	2.313e-02	–	3.710e-02	–	2.796e-01	–
$1/16$	6.377e-04	1.94	1.164e-03	2.00	6.481e-04	2.18	5.728e-03	2.01	4.628e-03	3.00	6.660e-02	2.07
$1/32$	1.680e-04	1.92	2.911e-04	2.00	1.547e-04	2.07	1.401e-03	2.03	5.784e-04	3.00	1.580e-02	2.08
$1/64$	4.260e-05	1.98	7.312e-05	1.99	3.792e-05	2.03	3.487e-04	2.01	7.999e-05	2.85	3.904e-03	2.02
$1/128$	1.076e-05	1.99	1.836e-05	1.99	9.436e-06	2.01	8.722e-05	2.00	1.438e-05	2.48	9.718e-04	2.01
$1/256$	2.704e-06	1.99	4.611e-06	1.99	2.359e-06	2.00	2.189e-05	1.99	3.209e-06	2.16	2.425e-04	2.00

Table 4. Free-slip, smooth forcing ($k = 8$).

h	$n = 2$				$n = 8$				$n = 32$			
	$E^{\mathrm{rel}}_{L_2}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2}(p)$	rate	$E^{\mathrm{rel}}_{L_2}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2}(p)$	rate	$E^{\mathrm{rel}}_{L_2}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2}(p)$	rate
$1/8$	3.794e-03	–	1.195e-02	–	3.584e-03	–	3.454e-02	–	4.638e-02	–	3.106e-01	–
$1/16$	6.230e-04	2.61	3.080e-03	1.96	7.176e-04	2.32	8.903e-03	1.96	5.447e-03	3.09	7.427e-02	2.06
$1/32$	1.320e-04	2.24	7.416e-04	2.05	1.772e-04	2.02	2.180e-03	2.03	6.411e-04	3.09	1.777e-02	2.06
$1/64$	3.650e-05	1.85	1.848e-04	2.00	4.601e-05	1.95	5.461e-04	2.00	9.064e-05	2.82	4.427e-03	2.00
$1/128$	9.792e-06	1.90	4.617e-05	2.00	1.177e-05	1.97	1.367e-04	2.00	1.858e-05	2.29	1.105e-03	2.00
$1/256$	2.524e-06	1.96	1.154e-05	2.00	3.023e-06	1.96	3.424e-05	2.00	4.520e-06	2.04	2.761e-04	2.00

Table 5. Zero-slip, delta-function forcing.

h	$n = 2$				$n = 8$				$n = 32$			
	$E^{\mathrm{rel}}_{L_2}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2}(p)$	rate	$E^{\mathrm{rel}}_{L_2}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2}(p)$	rate	$E^{\mathrm{rel}}_{L_2}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2}(p)$	rate
$1/8$	2.545e-01	–	2.832e-01	–	1.025e-01	–	3.449e-01	–	1.783e-01	–	8.167e-01	–
$1/16$	9.755e-02	1.38	2.019e-01	0.49	3.555e-02	1.53	2.473e-01	0.48	7.344e-02	1.28	6.355e-01	0.36
$1/32$	3.228e-02	1.60	1.388e-01	0.54	1.181e-02	1.59	1.707e-01	0.54	2.629e-02	1.48	4.544e-01	0.48
$1/64$	1.134e-02	1.51	9.797e-02	0.50	4.163e-03	1.50	1.205e-01	0.50	9.343e-03	1.49	3.228e-01	0.49
$1/128$	4.083e-03	1.47	6.972e-02	0.49	1.490e-03	1.48	8.563e-02	0.49	3.378e-03	1.47	2.302e-01	0.49
$1/256$	1.439e-03	1.50	4.924e-02	0.50	5.281e-04	1.50	6.061e-02	0.50	1.198e-03	1.50	1.630e-01	0.50

Table 6. Zero-slip, smooth forcing ($k = 2$).

h	$n = 2$				$n = 8$				$n = 32$			
	$E^{\mathrm{rel}}_{L_2}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2}(p)$	rate	$E^{\mathrm{rel}}_{L_2}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2}(p)$	rate	$E^{\mathrm{rel}}_{L_2}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2}(p)$	rate
$1/8$	5.169e-03	–	4.309e-03	–	4.692e-03	–	2.188e-02	–	4.621e-02	–	2.766e-01	–
$1/16$	1.287e-03	2.01	1.083e-03	1.99	9.929e-04	2.24	5.433e-03	2.01	6.648e-03	2.80	6.437e-02	2.10
$1/32$	3.253e-04	1.98	2.715e-04	2.00	2.375e-04	2.06	1.343e-03	2.02	8.624e-04	2.95	1.527e-02	2.08
$1/64$	8.137e-05	2.00	6.831e-05	1.99	5.873e-05	2.02	3.358e-04	2.00	1.206e-04	2.84	3.784e-03	2.01
$1/128$	2.041e-05	2.00	1.721e-05	1.99	1.467e-05	2.00	8.419e-05	2.00	2.004e-05	2.59	9.432e-04	2.00
$1/256$	5.103e-06	2.00	4.351e-06	1.98	3.662e-06	2.00	2.111e-05	2.00	4.126e-06	2.28	2.354e-04	2.00

Table 7. Zero-slip, smooth forcing ($k = 8$).

h	$n = 2$				$n = 8$				$n = 32$			
	$E^{\mathrm{rel}}_{L_2}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2}(p)$	rate	$E^{\mathrm{rel}}_{L_2}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2}(p)$	rate	$E^{\mathrm{rel}}_{L_2}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2}(p)$	rate
$1/8$	5.720e-03	–	9.905e-03	–	6.657e-03	–	2.663e-02	–	6.009e-02	–	2.829e-01	–
$1/16$	1.311e-03	2.13	2.585e-03	1.94	1.283e-03	2.38	6.863e-03	1.96	9.136e-03	2.72	6.628e-02	2.09
$1/32$	3.202e-04	2.03	6.338e-04	2.03	2.837e-04	2.18	1.696e-03	2.02	1.185e-03	2.95	1.590e-02	2.06
$1/64$	7.962e-05	2.01	1.596e-04	1.99	6.861e-05	2.05	4.264e-04	1.99	1.699e-04	2.80	3.969e-03	2.00
$1/128$	1.990e-05	2.00	4.003e-05	2.00	1.700e-05	2.01	1.069e-04	2.00	2.923e-05	2.54	9.910e-04	2.00
$1/256$	4.970e-06	2.00	1.003e-05	2.00	4.234e-06	2.01	2.680e-05	2.00	6.181e-06	2.24	2.476e-04	2.00

6 Discussion

The convergence behaviour separates primarily by forcing type. Smooth forcing preserves the regularity needed for second-order pressure convergence with the $P_2 \times P_1$ discretisation. Velocity convergence is also regular, although the observed rates are influenced by the curved geometry, boundary-condition enforcement, and the number of elements per azimuthal wavelength. These effects are most visible for $n = 32$, where the coarsest meshes under-resolve the angular structure.

The delta-function cases are more demanding because the forcing is concentrated on an internal circular interface. The slower convergence rates are consistent with the reference K2021 solutions and with the expectation that pressure is especially sensitive to reduced regularity in mixed Stokes discretisations. Importantly, the free-slip and zero-slip variants show the same qualitative rates, indicating that the UW3 boundary-condition implementations behave consistently across both benchmark families.

The benchmark is therefore a useful intermediate verification problem between Cartesian manufactured solutions and full spherical-shell mantle-convection calculations. It tests curved-domain Stokes assembly, pressure normalisation, boundary conditions, and both smooth and non-smooth buoyancy forcing while remaining compact enough for routine convergence testing.

7 Conclusions

We have presented a Copernicus-style summary of the UW3 Kramer annulus Stokes benchmark. The main findings are:

1. UW3 reproduces the analytical annulus solutions of K2021 for both free-slip and zero-slip velocity boundary conditions.
2. Smooth forcing cases show approximately second-order pressure convergence and systematic second-order or better velocity convergence over the tested mesh sequence.
3. Delta-function forcing cases show the expected reduced convergence, approximately $O(h^{1.5})$ in velocity and $O(h^{0.5})$ in pressure.
4. The observed distinction between smooth and interface-localised forcing reflects benchmark solution regularity and provides a useful test of finite element behaviour in curved geodynamic domains.

These results verify the UW3 implementation for annulus Stokes flow with analytical Kramer forcing and provide a compact reference for future annulus and spherical-shell benchmark comparisons.

References

Baumgardner, J. R.: Three-dimensional treatment of convective flow in the Earth's mantle, Journal of Statistical Physics, 39, 501–511, https://doi.org/10.1007/BF01008348, 1985.

Blankenbach, B., Busse, F., Christensen, U., Cserepes, L., Gunkel, D., Hansen, U., Jarvis, G., Koch, M., Marquart, G., Moore, D., Olson, P., Schmeling, H., and Schnaubelt, T.: A benchmark comparison for mantle convection codes, Geophysical Journal International, 98, 23–38, https://doi.org/10.1111/j.1365-246X.1989.tb05511.x, 1989.

Boffi, D., Brezzi, F., and Fortin, M.: Mixed Finite Element Methods and Applications, Springer Series in Computational Mathematics, Springer Berlin Heidelberg, ISBN 9783642365195, https://doi.org/10.1007/978-3-642-36519-5, 2013.

Busse, F. H., Christensen, U. R., Clever, R. M., Cserepes, L., Gable, C., Giannandrea, E., Guillou, L., Houseman, G., Nataf, H.-C., Ogawa, M., Parmentier, E., and Sotin, C.: 3D convection at infinite Prandtl number in Cartesian geometry: a benchmark comparison, Geophysical and Astrophysical Fluid Dynamics, 75, 39–59, https://doi.org/10.1080/03091929408203646, 1994.

Davies, D. R., Kramer, S. C., Ghelichkhan, S., and Gibson, A.: Towards automatic finite-element methods for geodynamics via Firedrake, Geoscientific Model Development, 15, 5127–5166, https://doi.org/10.5194/gmd-15-5127-2022, 2022.

Girault, V. and Raviart, P.-A.: Finite Element Methods for Navier–Stokes Equations, Springer Series in Computational Mathematics, Springer Berlin Heidelberg, ISBN 9783642616235, https://doi.org/10.1007/978-3-642-61623-5, 1986.

Jarvis, G. T.: Effects of curvature on two-dimensional models of mantle convection: cylindrical polar coordinates, Journal of Geophysical Research: Solid Earth, 98, 4477–4485, https://doi.org/10.1029/92JB02117, 1993.

King, S. D., Lee, C., van Keken, P. E., Leng, W., Zhong, S., Tan, E., Tosi, N., and Kameyama, M. C.: A community benchmark for 2-D Cartesian compressible convection in the Earth's mantle, Geophysical Journal International, 180, 73–87, https://doi.org/10.1111/j.1365-246X.2009.04413.x, 2010.

Kramer, S. C., Davies, D. R., and Wilson, C. R.: Analytical solutions for mantle flow in cylindrical and spherical shells, Geoscientific Model Development, 14, 1899–1919, https://doi.org/10.5194/gmd-14-1899-2021, 2021.

Moresi, L., Mansour, J., Giordani, J., Knepley, M., Knight, B., Graciosa, J. C., Gollapalli, T., Lu, N., and Beucher, R.: Underworld3: Mathematically Self-Describing Modelling in Python for Desktop, HPC and Cloud, Journal of Open Source Software, 10, 7831, https://doi.org/10.21105/joss.07831, 2025.

Tackley, P. J.: Modelling compressible mantle convection with large viscosity contrasts in a three-dimensional spherical shell using the yin-yang grid, Physics of the Earth and Planetary Interiors, 171, 7–18, https://doi.org/10.1016/j.pepi.2008.08.005, 2008.

Thieulot, C. and Bangerth, W.: On the choice of finite element for applications in geodynamics, Solid Earth, 13, 229–249, https://doi.org/10.5194/se-13-229-2022, 2022.

Thieulot, C. and Puckett, E. G.: Incompressible Stokes flow in an annulus: An analytical solution and numerical benchmark, https://www.math.ucdavis.edu/~egp/PUBLICATIONS/JOURNAL_ARTICLES/SUBMITTED/CAPT-EGP-2018.pdf, preprint submitted to Computers & Geosciences, 2018.

Tosi, N., Stein, C., Noack, L., Hüttig, C., Samuel, H., Maierová, P., Davies, D. R., Wilson, C. R., Kramer, S. C., Thieulot, C. A. P., Glerum, A. C., Fraters, M. R. T., Spakman, W., Rozel, A., and Tackley, P. J.: A community benchmark for viscoplastic thermal convection in a 2-D square box, Geochemistry, Geophysics, Geosystems, 16, 2175–2196, https://doi.org/10.1002/2015GC005807, 2015.

Zhong, S., Zuber, M. T., Moresi, L., and Gurnis, M.: Role of temperature-dependent viscosity and surface plates in spherical shell models of mantle convection, Journal of Geophysical Research: Solid Earth, 105, 11 063–11 082, https://doi.org/10.1029/2000JB900003, 2000.